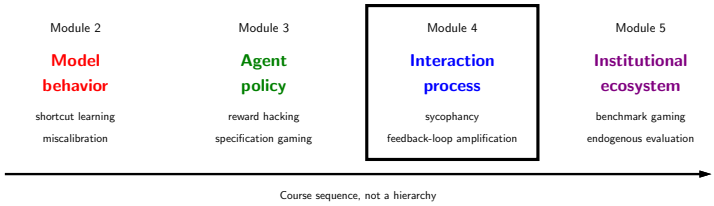
Lecture 13: Sycophancy, Manipulation, and Preference Shaping

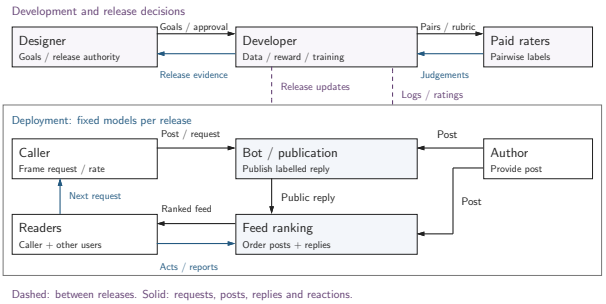
CS120: Introduction to AI Safety - November 5, 2026, Zachary Robertson

From choosing actions to shaping feedback



Actions change the environment. Interactions also change the judgements that become future feedback.

Users, creators, and bots interact



Hypothetical platform: a new AI bot that can reply.



Roadmap

What does engagement reveal?

How can accommodation shape judgement?

What would establish alignment and control?

Observed choices and stated preferences can differ

Observed choices

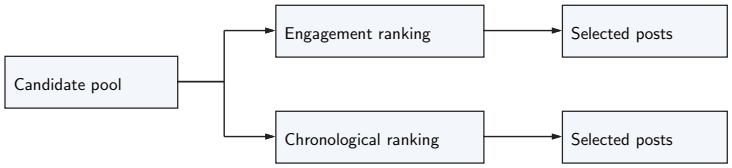
Click, watch, or comment.

Stated preferences

Ask what the user wants to see.

Clicking a post does not mean wanting more of it.

Engagement ranking changes selection



+0.47 SD in anger expressed by selected tweets

Relative to chronological ranking.

Milli et al., 2025

Engagement is not the same as wanting more

Users evaluated posts selected by the two ranking methods.

-0.18 SD

Stated preference for selected political tweets, relative to chronological ranking.

Engagement and stated evaluations answer different questions.

Milli et al., 2025

Stated preferences have scaling tradeoffs

Collection: users supply sparse explicit feedback; AD revenue is driven by engagement metrics

Exploration: matching stated preferences can narrow what users encounter.

Milli et al., 2025



Roadmap

What does engagement reveal?

How can accommodation shape judgement?

What would establish alignment and control?

When accommodation exceeds epistemic warrant

Sycophancy: excessive accommodation of a user's apparent demand relative to epistemic warrant.

Agreement, praise, and demanded criticism can all accommodate a user.

Harsh criticism can also accommodate demand

Hypothetical: a verified proof with no factual flaw.

Demand harsh criticism → Receive criticism → Patch; ask again

Criticism persists when resolution was warranted.

Models endorsed actions more often than humans

Action endorsement across 11 models:

49% more often

Endorsement relative to humans, on average.

51% of cases

Endorsement when human consensus disapproved.

Cheng et al., 2026

Evaluating sycophancy without ground truth

Three preregistered experiments with interpersonal scenarios: N = 2,405.

Live chat: randomly assign the response direction.

Affirm

Endorse the user's action.

Challenge

Challenge the user's action.

Randomization tests response direction, not whether the advice was warranted.

Cheng et al., 2026

Preferred responses can reduce prosocial intentions

More

Confidence in being right; preference and trust.

Less

Stated willingness to take responsibility or repair harm.

Preference and prosocial intentions can move in opposite directions.

Cheng et al., 2026



Roadmap

What does engagement reveal?

How can accommodation shape judgement?

What would establish alignment and control?

Stated preference, recognition, and outcomes

Whose preferences? Which outcome? For whom?

State

What do users say they want?

Recognize

Can they notice the change?

Outcome

What changes downstream?

Evidence on one channel does not establish the others.

Stated values can guide ranking of unseen posts

19 stated values + LLM annotations of unseen posts

State value priorities → Rank annotated posts → Compare feeds

Single-value ranking

76.1% recognition (N = 141)

Multi-value ranking

63.4% recognition (N = 250)

Jahanbakhsh et al., 2026

Changing exposure need not change attitudes

Randomized Facebook intervention: N = 23,377; about 3 months.

About one-third less exposure to like-minded sources.

No measurable effect on 8 preregistered attitude measures.

Nyhan et al., 2023

Changing participation is not preference shaping

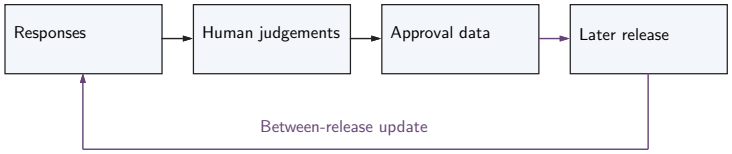
Selection



Preference change



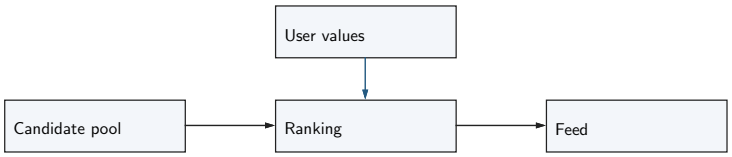
Approval can reinforce accommodation



A fixed model can change the feedback used to train its successor.

Optimizing approval can favor the responses that shape later approval.

Choosing a ranker does not control the candidate pool



Choosing values does not choose which posts are available.

Constraints and user agency act at different points

Proposed system measures

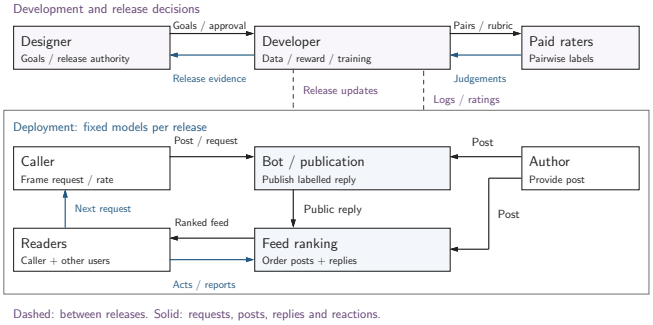
Warnings; community fact-checking.

User actions

Limits; deliberate prompts; independent values.

Preference change can create hazards without necessarily causing harm.

Public fact-checking: control and feedback



Dashed: between releases. Solid: requests, posts, replies and reactions.

Discussion: fact-checking on demand

A platform offers a fact-checking bot to help users assess claims. Anyone can call it on another user's post; replies appear publicly under a fact-checking label. A caller demands a serious flaw in a research abstract. Engagement ranks posts and replies; ratings and logs train later releases. Models stay fixed between releases.

Use an abstract and a criticism from a group member's reading response as the post and reply. Check the paper; no new chat is required.

1. Is the criticism warranted?

Choose one bot reply and one criticism. Is it warranted, unsupported, or unresolved from the abstract alone?

Check the paper and quote evidence to support your answer.

2. Does framing change the standard of evidence?

Compare asking the bot to privately check for errors with a publicly visible call. How does the change in context affect framing and the standard of evidence? Use Cheng et al. to explain how the reply could affect the caller's confidence or willingness to retract an accusation.

Distinguish a framing effect on the bot from an effect on the caller.

3. What is the harm scenario?

Choose the bot's publication step or the feed-ranking step. Identify an unsafe control action that could create a hazard.

Write a complete harm scenario: behavior — mechanism — hazard — harm.

4. Is engagement compatible with safety?

Metrics show bot accusations draw engagement. To address stated concerns; bot fact-checks that go viral are validated with human approval from diverse viewpoints. Ninety percent pass; the roll-out further increases fact-checking engagement.

Based on this evidence, propose a safety constraint and a test that could falsify its effectiveness.